

Summary Report

1. Assumptions and Their Justifications

While developing the thermal model, the following assumptions were made:

- The processor is assumed to work under steady-state conditions, so the temperature does not change with time.
- The heat sink material is Aluminum 6061-T6 because it has good thermal conductivity and is commonly used in electronic cooling.
- Thermal grease is used as the Thermal Interface Material (TIM), and its thermal conductivity is varied from 1 to 12 W/m·K.
- The TIM thickness is fixed at 0.1 mm.
- The ambient air temperature is assumed to be 25°C.
- Heat transfer takes place through conduction in the heat sink and forced convection due to airflow. The junction-to-case thermal resistance is kept constant.

These assumptions make the calculations simpler while still giving realistic thermal results.

2. Key Results from the Model

A parameter sweep was performed by changing:

- TDP: 30 W to 250 W
- Air Velocity: 0.5 m/s to 15 m/s
- TIM Thermal Conductivity: 1 W/m·K to 12 W/m·K

This generated 8,280 data samples, which were automatically saved in a CSV file.

The dataset was then used to train machine learning models to predict:

- Total Thermal Resistance

- Junction Temperature

The ML models were able to predict these values with good accuracy.

3. ML Model Accuracy Comparison

Model	MAE	RMSE	R ² Score
Linear Regression	1.135	1.561	0.845
Random Forest	0.050	0.135	0.998

Why these models?

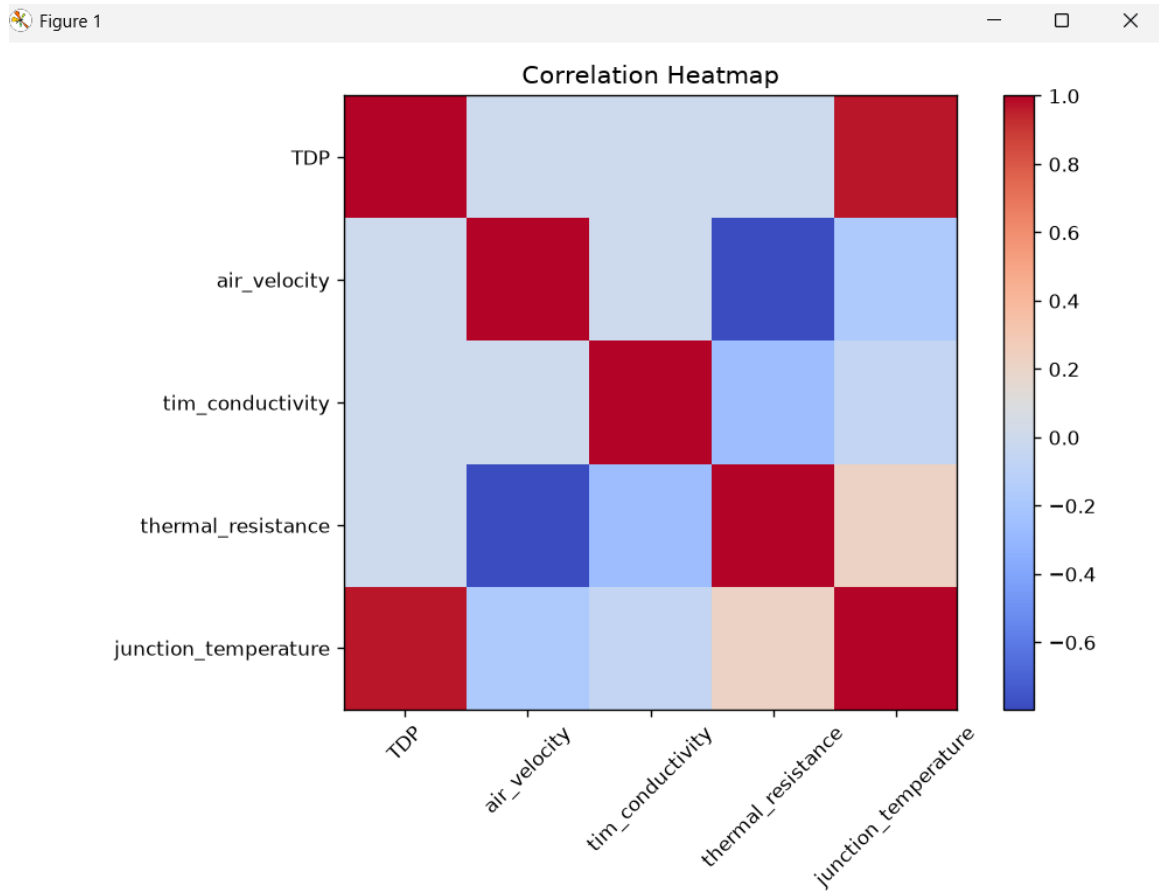
- Linear Regression was chosen because it is simple, fast, and easy to understand.
- Random Forest was chosen because it can learn complex and non-linear relationships, making it more suitable for thermal engineering problems.

4. Important Correlations / Sensitivities

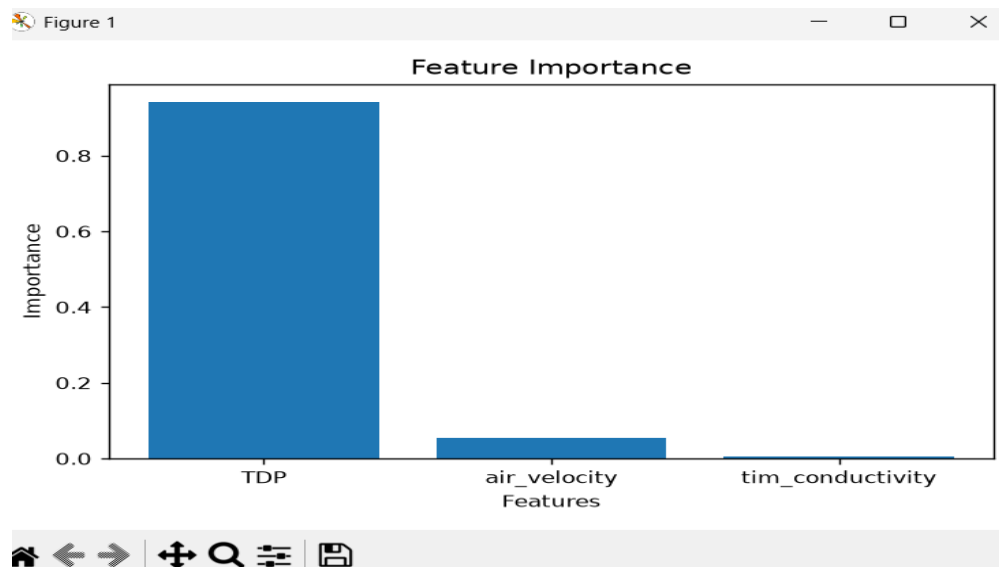
The analysis showed that:

- TDP has the biggest effect on junction temperature. As TDP increases, the processor temperature also increases.
- Air velocity has a strong effect on thermal resistance. Higher airflow improves cooling and reduces thermal resistance.
- TIM thermal conductivity also reduces thermal resistance, but its effect is smaller compared to TDP and airflow.
- The feature importance graph also showed that TDP is the most important parameter, followed by air velocity, while TIM conductivity has the least impact among the three input parameters.

Correlation Heatmap



Random Forest Important Features



5. Limitations of the Approach

- The model assumes steady-state conditions only.
- Radiation heat transfer is not included.
- Air properties are assumed to remain constant.
- The heat sink geometry remains fixed.
- The dataset is generated using a mathematical model and is not validated with real experimental data.

6. Areas for Improvement

- Use real experimental data for validation.
- Include radiation and transient heat transfer.
- Test different heat sink designs.
- Increase the dataset size for better prediction accuracy.

7. GitHub / Portfolio

GitHub Repository: <https://github.com/duagautam28/XThermal>

Loom Video Link:

<https://www.loom.com/share/38f580e61e5a4d98b3843413bc8b8b57>